

LEARNER WORKBOOK

NAME: _____

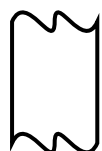
YEAR 3 _____

TERM 1



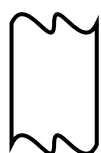
General Safety in the Workshop

The Woodwork learners work with hand tools, machinery and other equipment in the Woodwork Centre. Before any work can be started, learners should know the basic principles regarding safe-work-practices to ensure that no accidents will take place.



What is an accident?

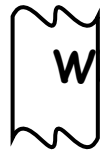
An accident is an unplanned and uncontrolled event, caused by unsafe acts or unsafe conditions



What is an unsafe act?

The word "act" refers to the word action (something we do), such as:

1. work without permission or work when teacher is not in class
2. fool around, tease others, run and play or fight in the Workshop.
3. use equipment unsafely or use unsafe or broken equipment.
4. adjust machinery or equipment whilst it is moving.
5. act irresponsibly: taking chances.
6. take up unsafe positions or stand too close to learners working on machinery
7. not using safety equipment or failure to wear personal protective equipment (PPE)
8. disturb learners whilst working on machinery and equipment
9. work when you are not feeling well or are under the influence of illegal substance and alcohol
10. tamper with safety equipment (e.g. fire extinguisher)
11. stack equipment unsafely - on top of one another



What is an unsafe condition?



The word "condition" refer to a situation, such as:

1. no guards on machinery, conveyors or belt drives,
2. defective working conditions; such as rough, sharp or slippery surfaces,
3. not using personal protective equipment or PPE is not available,
4. lighting and ventilation in workshop is poor,
5. workshop is too overcrowded,
6. workshop is dirty and untidy - waste and material on the floor,
7. flammable chemicals and substances are stored near heat or sunlight,
8. placing spanners, bolts, nuts and wood on equipment while working,
9. electrical cables are not insulated or secured or lying on floor,
10. machinery and equipment are not supplied with emergency switches.

WEEK 1	QUESTIONS	General Safety in the Centre
Activity sheet: Safety		

The Woodwork learners work with hand tools, machinery and other equipment in the Woodwork Centre. Before any work can be started, learners should know the basic principles regarding safe-work-practices to ensure that no accidents will take place.

What is an accident? An accident is an _____ and _____ event, caused by unsafe acts or unsafe conditions

What is an unsafe act? The word "act" refers to the word _____ (something we do)

1. work without _____ or work when teacher is not in class,
2. fool around, tease others, _____ and _____ or fight in the workshop,
3. use equipment unsafely or use _____ or broken equipment,
4. adjust machinery or equipment whilst it is _____,
5. _____ irresponsibly: taking chances,
6. take up unsafe positions or stand too _____ to learners working on machinery
7. not using _____ equipment or failure to wear personal protective equipment
8. _____ learners whilst working on machinery and equipment,
9. work when you are not _____ well or are under the influence of illegal substance and alcohol,
10. _____ with safety equipment (e.g. fire extinguisher)
11. _____ equipment unsafely - on top of one another

What is an unsafe conditions? The word "condition" refer to a situation.

1. no _____ on machinery, conveyors or belt drives,
2. defective _____ conditions; such as rough, sharp or slippery surfaces.
3. not using personal _____ equipment or is not available



4. lighting and _____ in workshop is poor,
5. workshop is too _____
6. workshop is dirty and untidy - _____ and material on the floor
7. flammable chemicals and substances are _____ near heat or sunlight
8. placing _____, bolts, nuts and wood on equipment while working
9. electrical _____ are not insulated or secured
10. machinery and equipment are not supplied with emergency

SAFETY

PPE

PPE Personal Protective equipment

1. Protection for the head: Helmet
2. Protection for the eyes; Goggles
3. Protection for breathing: a Mask
4. Protection for the ears: Ear muffs
5. Protection for the body: Overall
6. Protection for the Hands: Gloves
7. Protection for the feet: Boots

Name the PPE?



NAME: _____

ACTIVITY 1

	What is my name? _____	What am I used for? _____ _____ _____ _____ _____ _____
	What is my name? _____	What am I used for? _____ _____ _____ _____ _____ _____
	What is my name? _____	What am I used for? _____ _____ _____ _____ _____ _____
	What is my name? _____	What am I used for? _____ _____ _____ _____ _____ _____



What is my name?

What am I used for?



What is my name?

What am I used for?



What is my name?

What am I used for?

SOLVENTS

1. **SOLVENTS** are liquids that give off **FUMES**.
2. Some fumes are dangerous and others can make you high.
3. Make sure there is enough FRESH AIR circulating.
4. Do not inhale solvents through your mouth or nose.
5. Keep solvents away from HEAT to prevent an explosion.
6. Store in a safe place.
7. Wear a mask when working with solvents
8. examples of solvents are: paint, thinners, paraffin, petrol and turpentine.



ACTIVITY

Are solvents flammable? _____

Fumes are _____ and can make you _____

Make sure there is enough _____

Store in a _____ place.

Wear a _____ when working with solvents

HIV AIDS

How do we get AIDS?

Through body fluids, blood transfusion, sex and breast milk.

What does HIV do?

It kills the white blood cells and destroys the immune system .

Can it be treated?

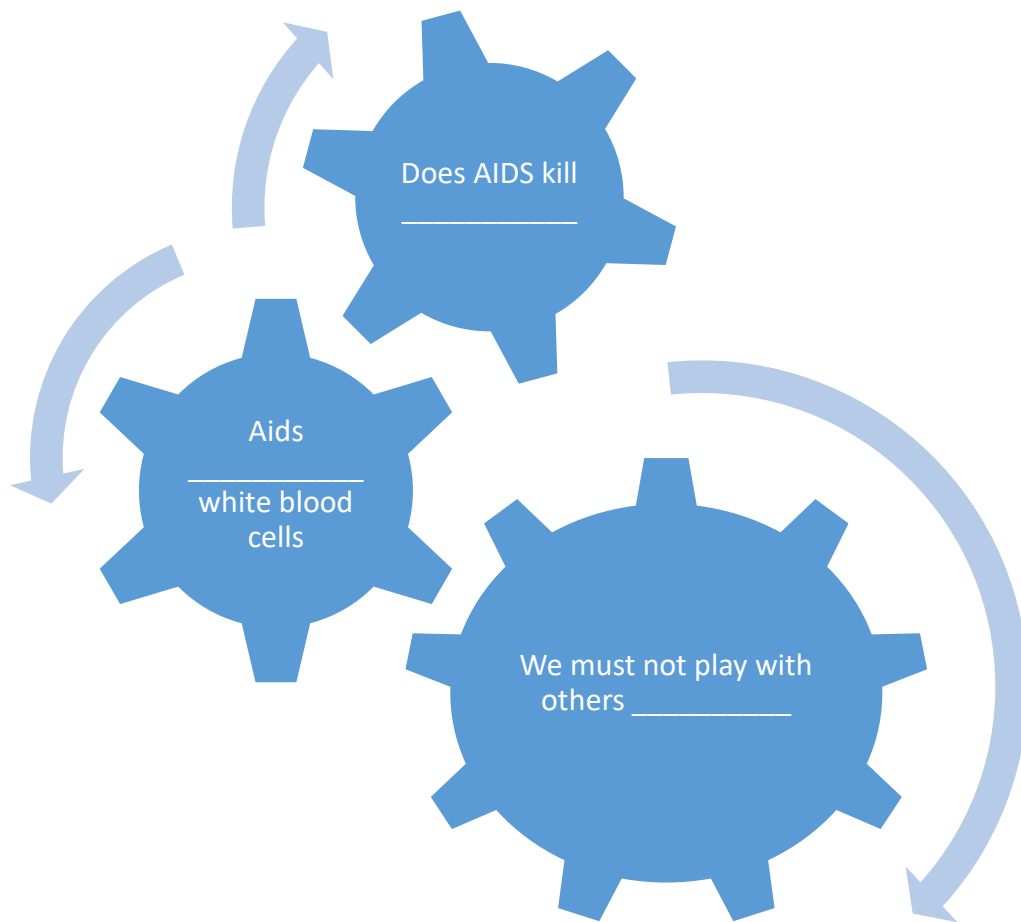
Yes

Will it kill you?

YES

WARNING!!!! DO NOT PLAY WITH ANOTHERS BLOOD OR BODY FLUIDS

ACTIVITY



EQUIPMENT

MACHINERY

SURFACE PLANER AND THICKNESSER

***Is used to make wood surfaces smooth and reduce the thickness of wood
It is also used to make wood even in thickness***

SAFETY when using a Surface Planer:

- ***Always wear eye and ear protection.***
 - Wearing a dust mask
 - Do not wear loose clothing or gloves.
 - Remove watches, all jewellery, and tie back long hair or beards
 - Ensure that the machine does not move during operation
 - . Wood pieces should be securely clamped before machining begins
 - Are push sticks or blocks used to move the timber through the machine.
 - Check if guards are secure and fitted
 - Are the stop/start buttons, emergency stop buttons, and safety bars working
 - Is there a clear workspace around the machine.
 - Is dust extraction fitted correctly.
 - Wear correct PPE
-



CIRCULAR SAW



The Circular saw is used for making long straight cuts.

- It can also be used for cutting at various angles

PARTS OF THE CIRCULAR SAW

ON & OFF SWITCH to switch machine on and off

HANDLE to hold and guide saw

CORD AND PLUG To provide power

BEVEL ADJUSTMENT to adjust the angle

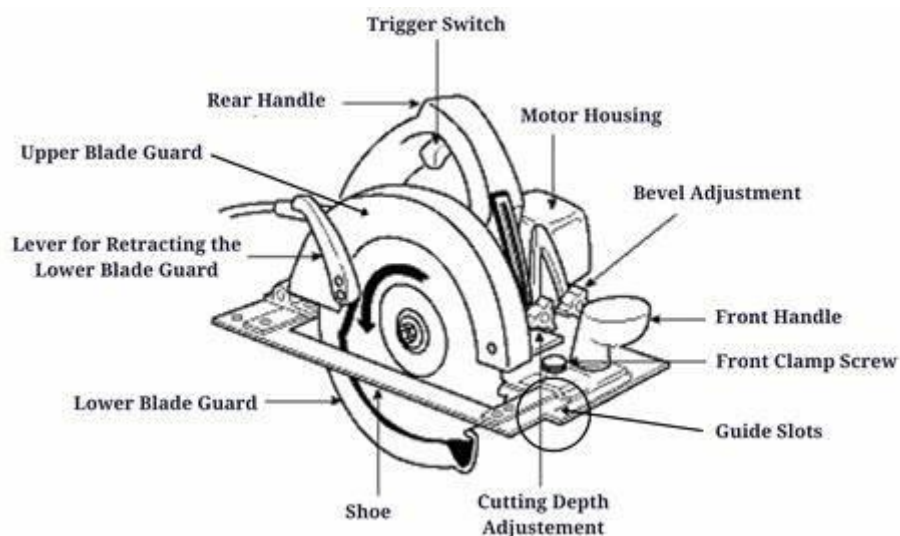
DEPTH ADJUSTMENT to adjust how deep the cut will be

BLADE for cutting the wood

BLADE COVER to protect you from getting cut

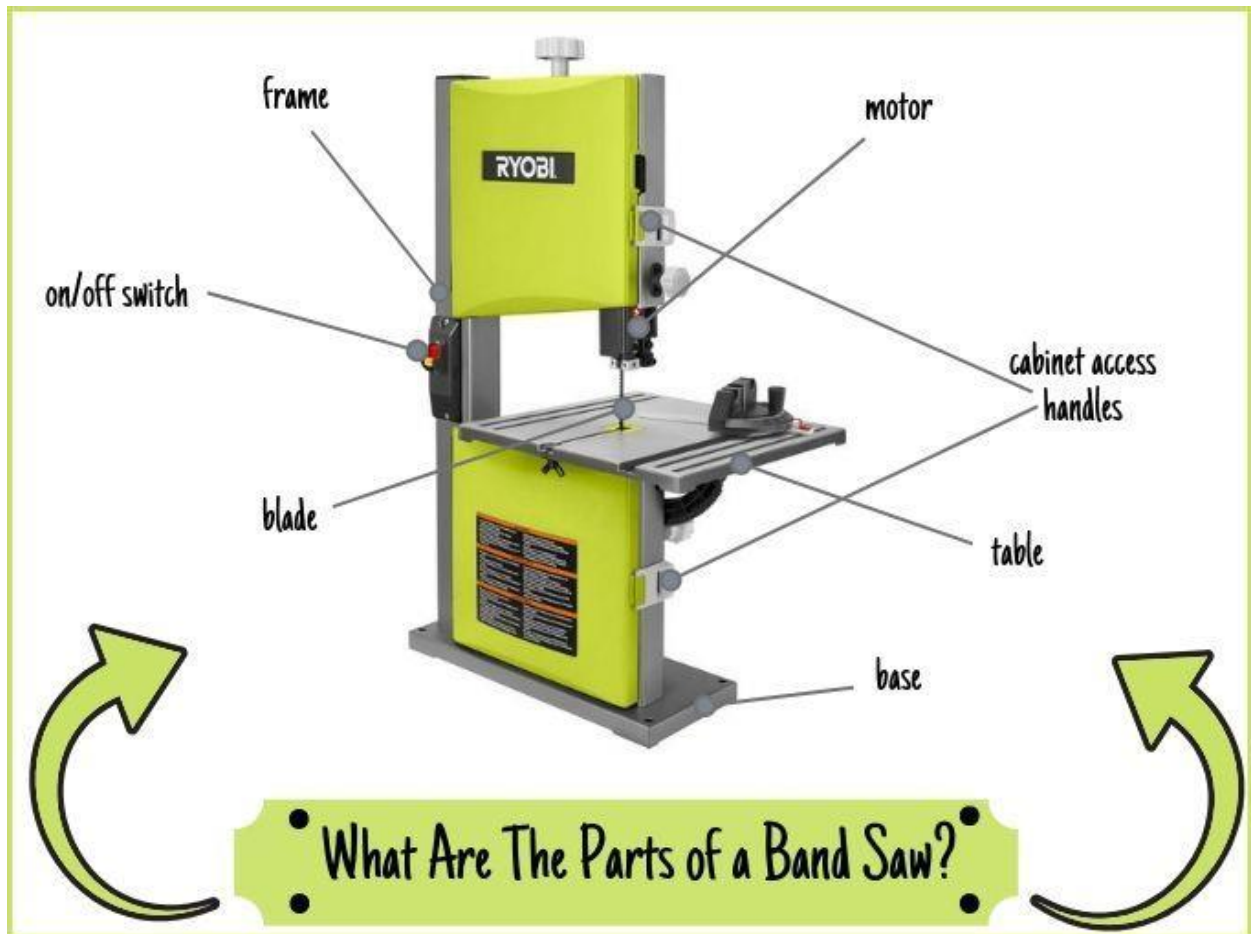
GUARD LEVER to keep the blade covered as the blade cuts deeper

BASE OR SHOE allows you to rest the saw on the material and to guide your cut



BAND SAW

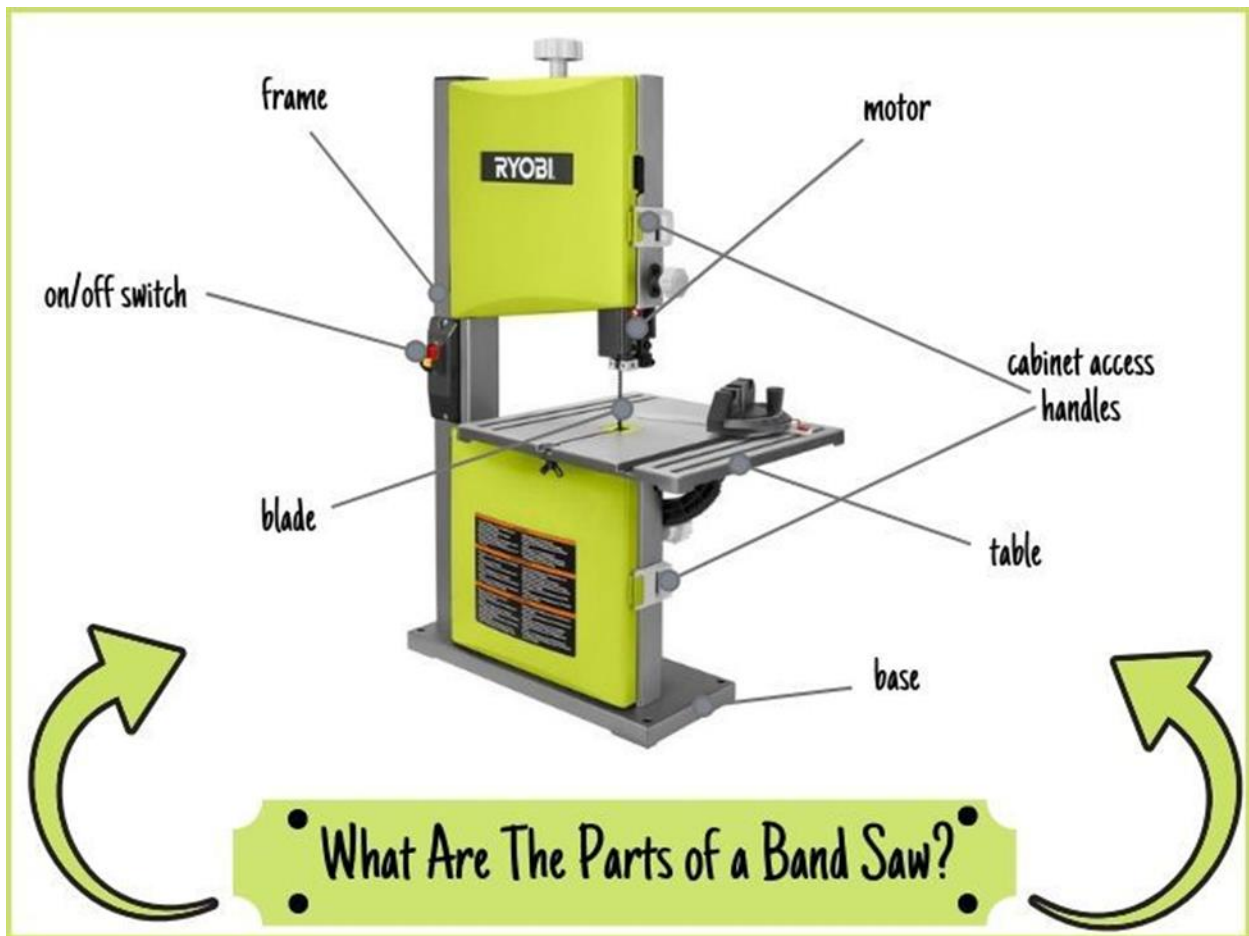
- **Resawing logs or lumber into thinner pieces or different sizes**¹²
- Cutting curves and intricate designs for detailed woodworking projects¹²
- Creating veneers and thin slices of wood or other materials²
- Craftsmanship and joinery, such as making tenons, dovetails, and lap joints²
- Metal cutting and shaping, such as cutting and shaping metal sheets and tubes



Safety Procedures

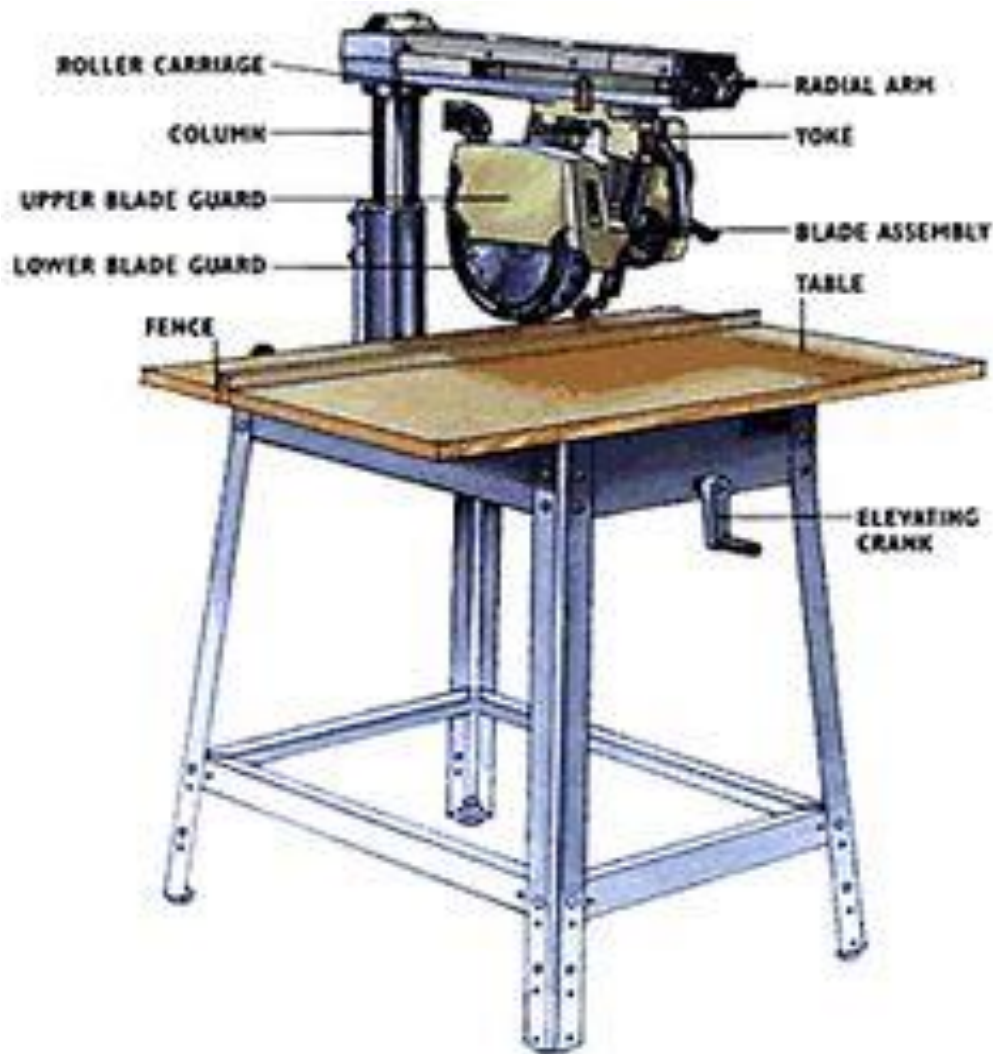
- • Wear proper Safety Gear:
 - Wear gloves, protective glasses, noise protection earmuffs, and an apron or smock.
- Inspect Blade Condition:
 - Make sure that blades are securely fastened and not damaged in any way.
- Keep Hands Away from the Blade:
 - Stand at a Safe Distance:
 - Unplug the Machine After Use:
- Clean area after working

ACTIVITY



Which machine do we use to cut curves

Radial arm saw



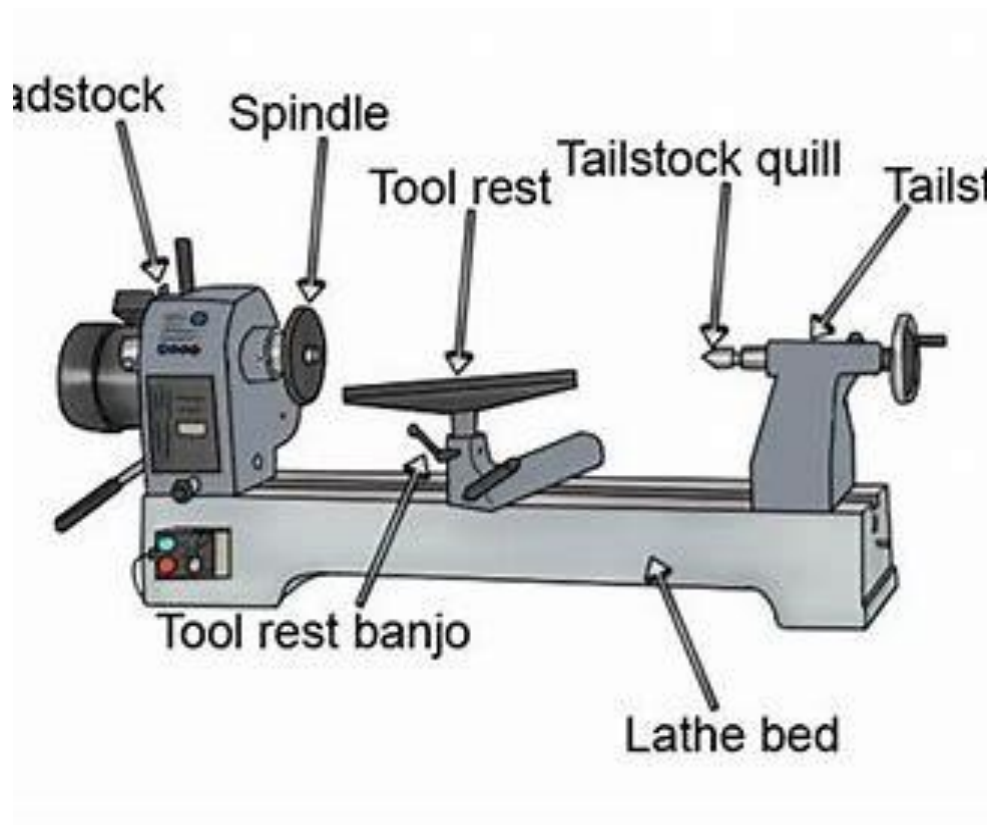
How to operate a band saw safely?

- Keep hands away from the blade and cutting area.
- Feed the work piece forward evenly and hold it firmly keeping hands in a safe position.
- Use a push stick when feeding material past the blade.
- Before making adjustments, switch off the machine.
- Wear safety gear such as gloves, protective glasses, noise protection earmuffs, and an overall.

WOOD LATHE

Lathe safety guidelines before using

- Inspect and adjust all safety devices.
- Double-check the guards are secure.
- Ensure that the machine has a start/stop button that works.
- Ensure that all machines have an emergency stop button (e-stop).
- Before turning on the power, make sure all obstacles are removed.
- Make sure all blades and cutting instruments are clean and sharp.
- Before measuring, cleaning, or making any modifications, turn off the machine
- Keep your hands clear of the cutting head and any other moving parts.
- Avoid awkward hand positions and operations.
- After each use, clean all tools and machine.
- Maintain a clean, well-swept, and well-lit work environment.
- Ensure there is enough space around the machine.

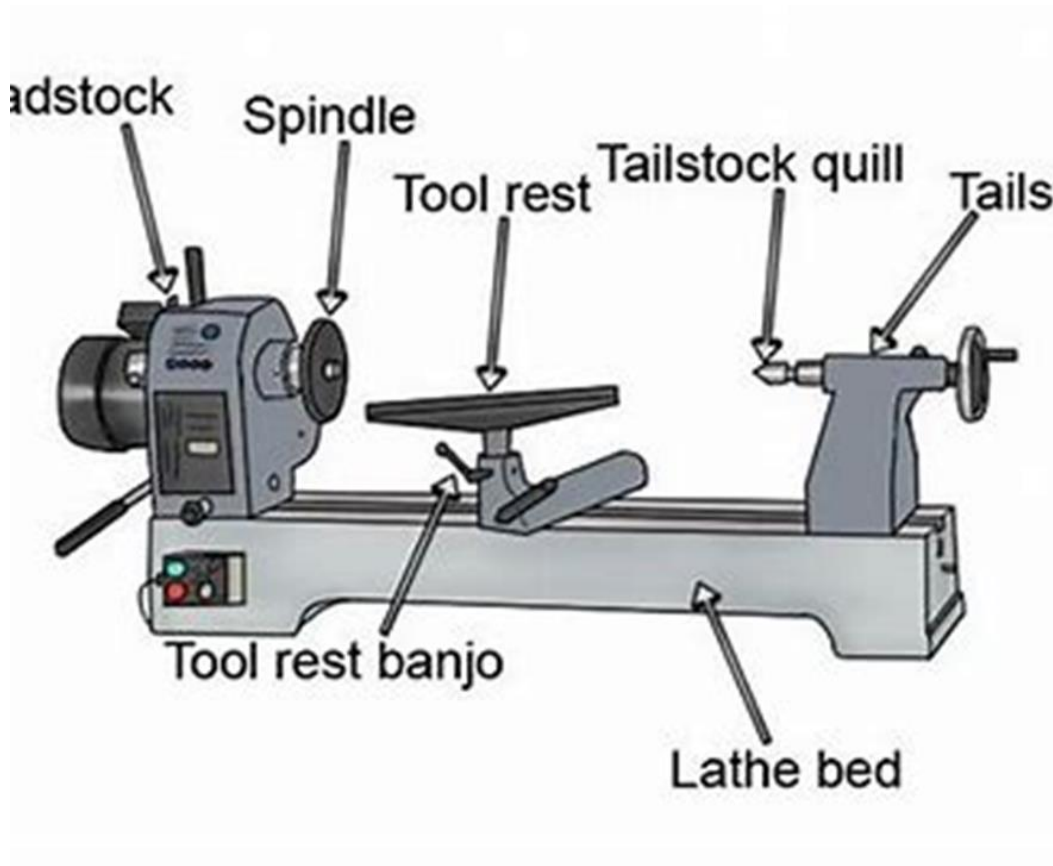


ACTIVITY

Fill in the missing words:

- _____ and adjust all safety devices.
- Double-check the _____ are secure.
- Ensure that the machine has a start/stop _____ that works.
- Ensure that all machines have an emergency _____ button (e-stop).
- Before turning _____ the power, make sure all obstacles are removed.
- Make sure all _____ and cutting instruments are clean and sharp.
- Before measuring, cleaning, or making any modifications, turn _____ the machine
- Keep your hands clear of the cutting head and _____ other moving parts.

- Avoid awkward _____ positions and operations.
- After each use, _____ all tools and machine.
- Maintain a clean, well-swept, and well-lit _____ environment.
- Ensure there is enough _____ around the machine



WOODWORK

TERM 2

YEAR 3



LEARNER NAME: _____

YEAR: _____

Information sheet: - Identification of electrical tools

Electrical tools	Name	Uses
	Circular saw	• The Circular saw is used for making long straight cuts. • It can also be used for cutting at various angles



Drill

- The drill is used for making holes in work surfaces, e.g. wood, steel or masonry
- When drilling into masonry, the hammer function is used



Jig saw

- It is used for cutting curves, patterns and shapes
- It can also be used for easy cutting in small spaces



Router

- It is used for shaping the edges of wood
- It can also be used for trimming edges, engraving and moulding work pieces.



Mitre saw

- It is used for cross cutting small wood
- It can also be used for cutting various angles on wood (called mitre cutting)

Activity sheet: - Identification of electrical tools

Electrical tools	Name	Uses
	_____ saw	• It is used for making long _____ cuts. • It can also be used for cutting at various _____



- It is used for making _____ in work surfaces, e.g. wood, steel or masonry
- When drilling into masonry, the hammer function is used



_____ saw

- It is used for cutting curves, _____ and shapes
- It can also be used for easy cutting in small spaces



- It is used for shaping the _____ of wood
- It can also be used for trimming edges, engraving and moulding work pieces.



_____ saw

- It is used for cross cutting small wood
- It can also be used for cutting various _____ on wood (called mitre cutting)



screwdriver

- It is used to _____
And _____ screws
fast

Day 28	Lesson 16	TOOLS
Information sheet: Identification of machinery		

Machinery	Name	Uses
-----------	------	------



Bench grinder

- The Bench grinder is used for tapering (file) steel objects
- It can also be used for sharpening and removing excess material



Table saw

- The Table saw is used to rip, miter, shape, square, groove, and join.
- It is also used for cutting larger work pieces, or when long straight cuts are required



Vertical Band saw

- The Band saw for cutting precise and complex shapes and curves



Radial arm saw

- The radial arm saw is used to perform crosscuts.
- It can also be used for cutting bevels and mitres, rabbets, mouldings, and even as a router guide



Surface planer

- The Surface planer is used for removing excess stock of boards and wood
- It can also be used to smooth out or plane surfaces evenly



Combination: Belt & Disk sander

- The Combination Belt & Disk sander is used for aggressive sanding of surfaces.



Drill press

- The Drill press is used for precision drilling and deliver large accurate diameter holes
- It can also be used for setting depths of holes and various types of drill bits

Day 28	Lesson 16	TOOLS
Activity sheet: Identification of machinery		

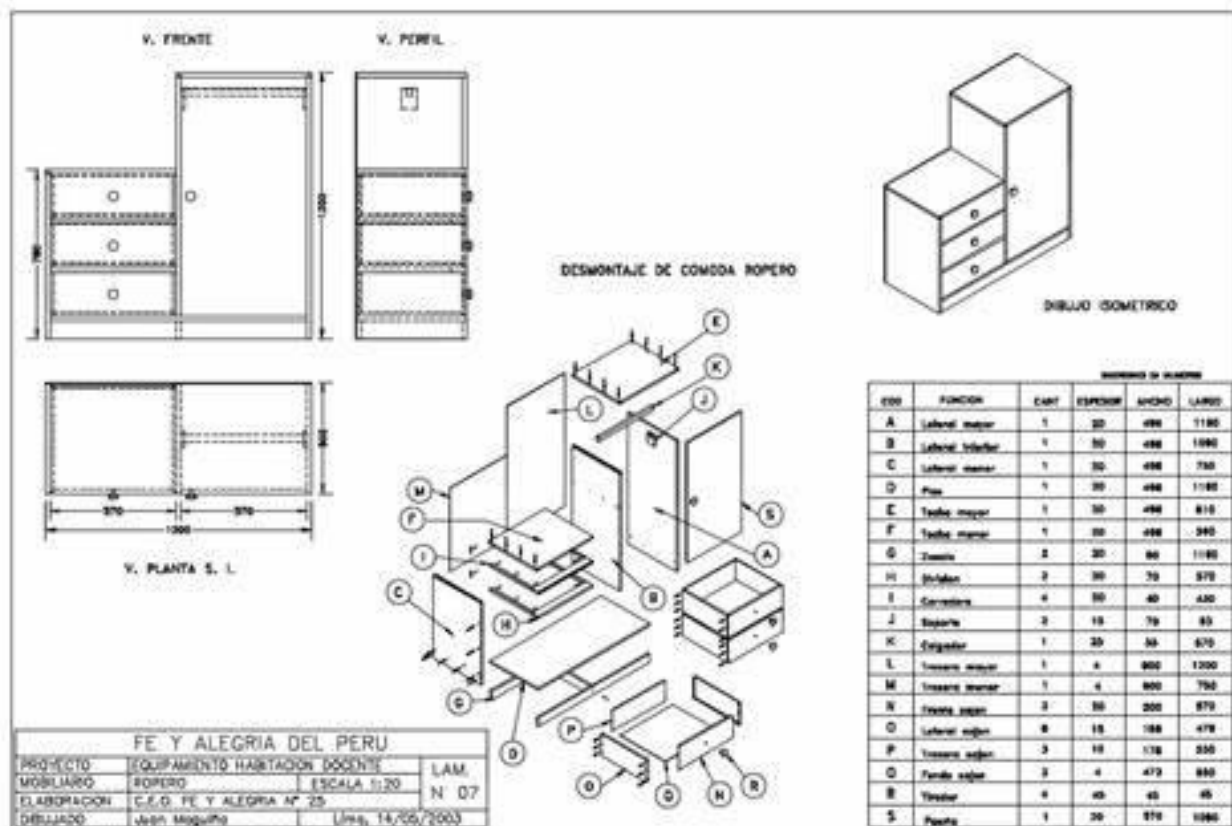
Machinery	Name	Uses
	_____ grinder	• It is used for _____ (file) steel objects • It can also be used for sharpening and _____ access material
	_____	• It is used to _____, miter, _____, square, _____, and join. • It is also used for cutting _____ work pieces, or when long straight cuts are required
	Vertical _____ saw	• It is used for cutting precise and complex _____ and _____
	_____ arm saw	• It is used to perform _____. • It can also be used for cutting bevels and mitres, rabbets, mouldings, and even as a router guide
	Surface _____	• It is used for removing _____ stock on boards and wood • It can also be used to smooth out or plane surfaces evenly
	Combination: _____ & _____ sander	• It is used for aggressive _____ of surfaces.
	Drill _____	• It is used for precision _____ and deliver large accurate diameter holes • It can also be used for setting _____ of holes and various types of drill bits

	Biscuit joiner	It is used to cut grooves for biscuits to join two pieces of wood
	Orbital sanders	Creating ultra-smooth surfaces on materials such as wood, plastic, and metal.
	_____ joiner	It is used to _____ grooves for biscuits to _____ two pieces of wood
	Orbital _____	Creating ultra-smooth _____ on _____ such as wood, plastic, and metal.

ACTIVITY

WOODWORK

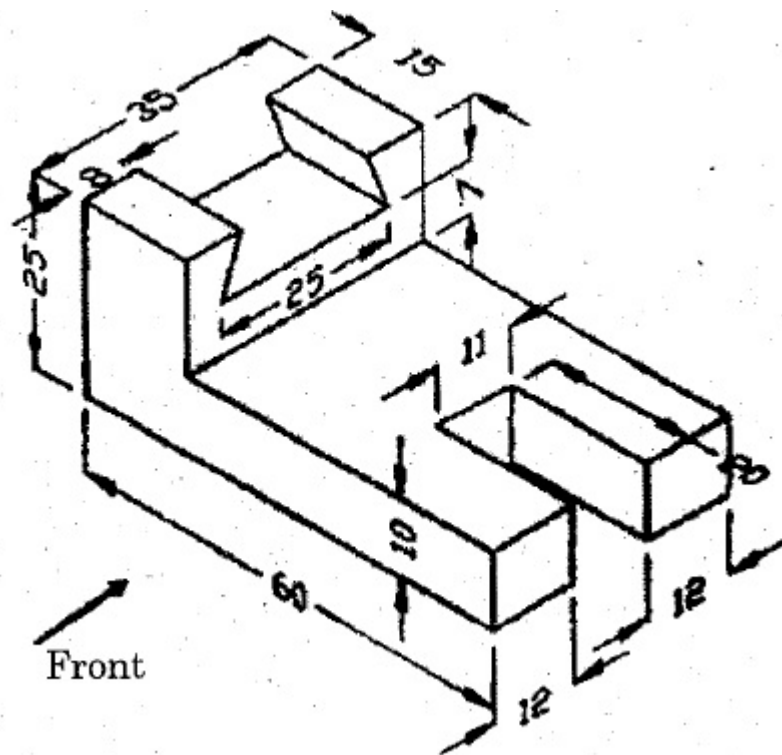
DRAWINGS



You will be doing two types of drawings:

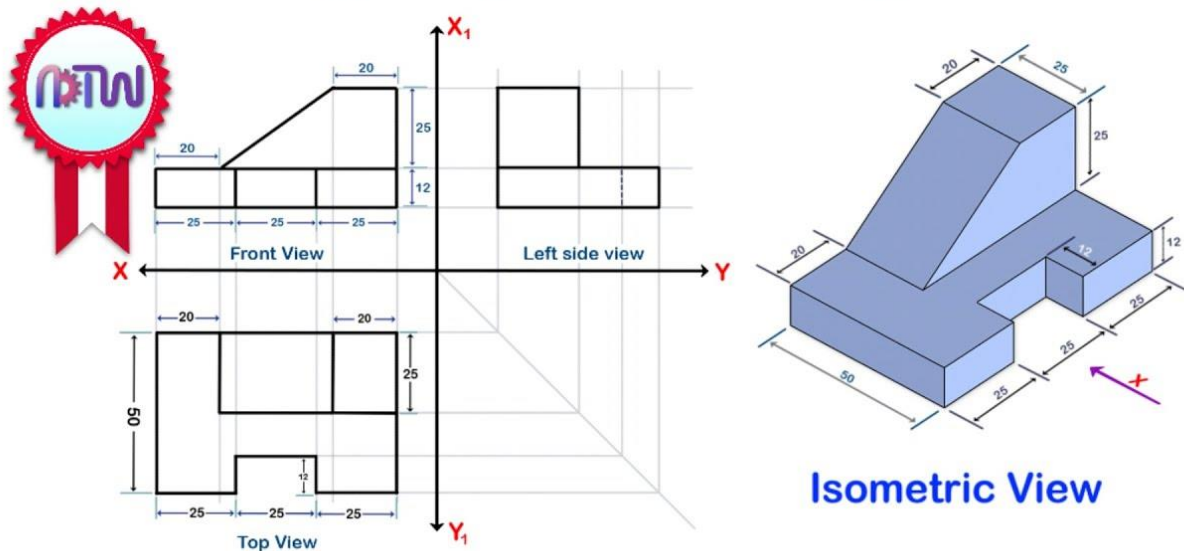
- ISOMETRIC DRAWINGS
- ORTHOGRAPHIC PROJECTIONS

ISOMETRIC DRAWINGS



ORTHOGRAPHIC PROJECTIONS

Orthographic Projection

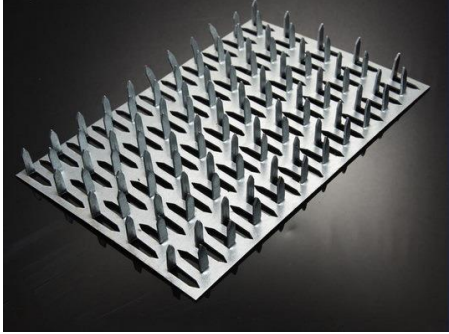


○ FASTENINGS



- Identify and know the different uses of the following fastening consumables
 - Corrugated fasteners
 - Nail plates
 - Expansion bolts
 - Lamello biscuits
 - Dowels
 - Wall plugs
 - Hammer-in fixing

PICTURE	WHAT AM I	WHAT AM I USED FOR
---------	-----------	--------------------

	Corrugated fastener	I join 2 pieces of wood when making frames
	Nail plate	I join wood when making roof trusses
	Expansion bolts my other name is Rawl bolts	I fasten wood and metal beams into walls
	Lamello biscuit	I fasten cabinet panels together



Dowels

I join wood in furniture making



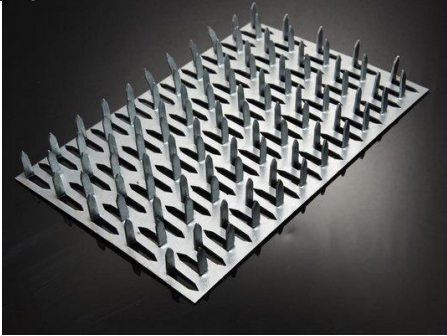
Wall plugs

I help screws to fix wood



**Hammer in fixing
I have a plastic plug
and a screw like nail
that gets hammered into
the wall**

**I am used to fix wood
against walls**

PICTURE	WHAT AM I	WHAT AM I USED FOR
	_____ fastener	I join 2 pieces of wood when making _____
	_____ _____	I join wood when making roof t _____
	Expansion bolts my other name is _____ _____	I fasten wood and metal beams into _____
	Lamello _____	I fasten _____ panels together



I join wood in
_____ making



I help screws to fix
wood



Hammer in fixing
I have a _____
plug and a screw like nail
that gets _____
into the wall

I am used to fix
wood against walls

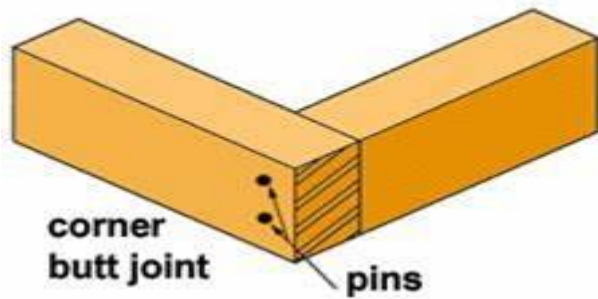
JOINING



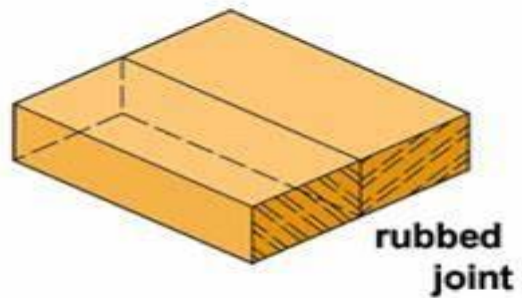
- Identify and make all the joints done during year 1 and 2 plus the following Joints to widen boards
 - Butt or Rubbed joint
 - Tongue-and-groove
 - Loose-tongue joint
 - Dowelled joint
 - Lamello biscuit joint

BUTT OR RUBBED JOINT

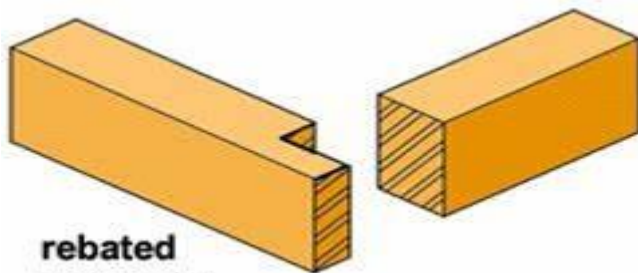
BUTT JOINTS



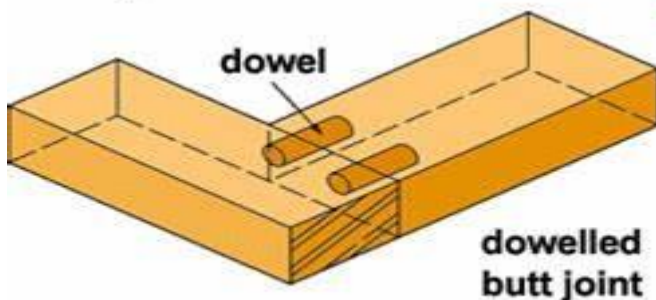
corner butt joint



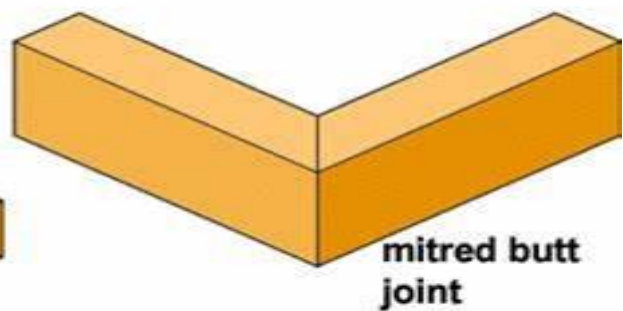
rubbed joint



rebated butt joint



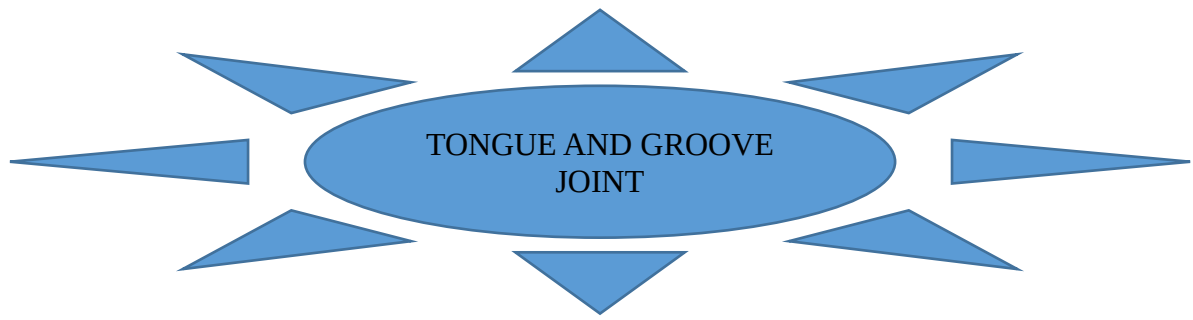
dowelled butt joint



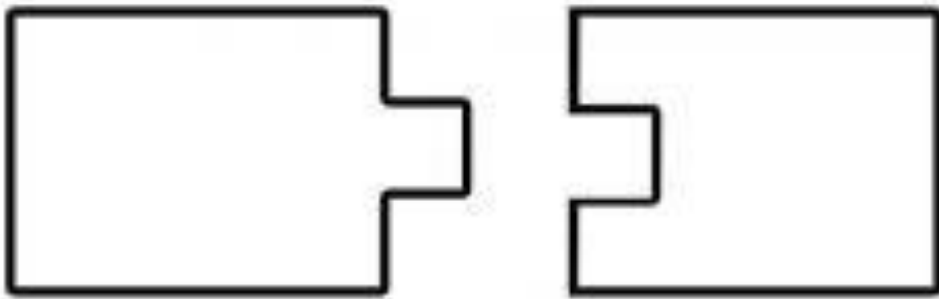
mitred butt joint

copyright Donal O'hEanaigh 2000

These joints are fixed together by using a fastener



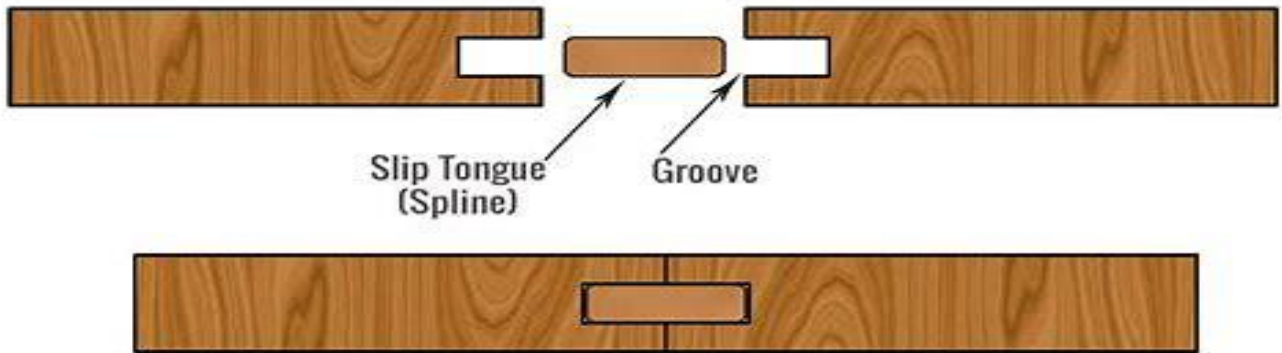
TONGUE AND GROOVE JOINT



ONE PIECE OF WOOD IS CUT WITH A TONGUE
AND ANOTHER IS CUT WITH A GROOVE AND
SLIDE TOGETHER TO FORM THE WOOD JOINT

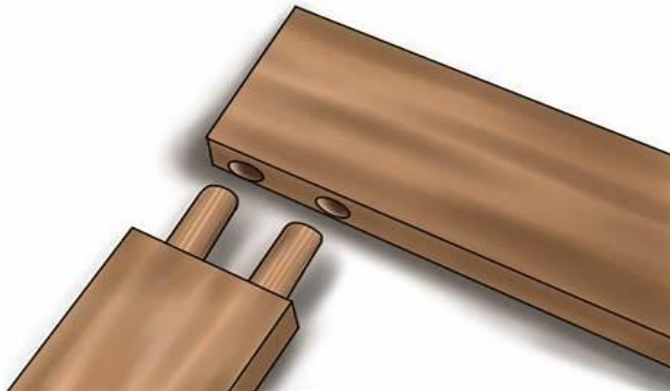
TYPES OF WOOD JOINTS & THEIR USES @BUILDEROLOGY.COM

SLIP AND TONGUE GROOVE

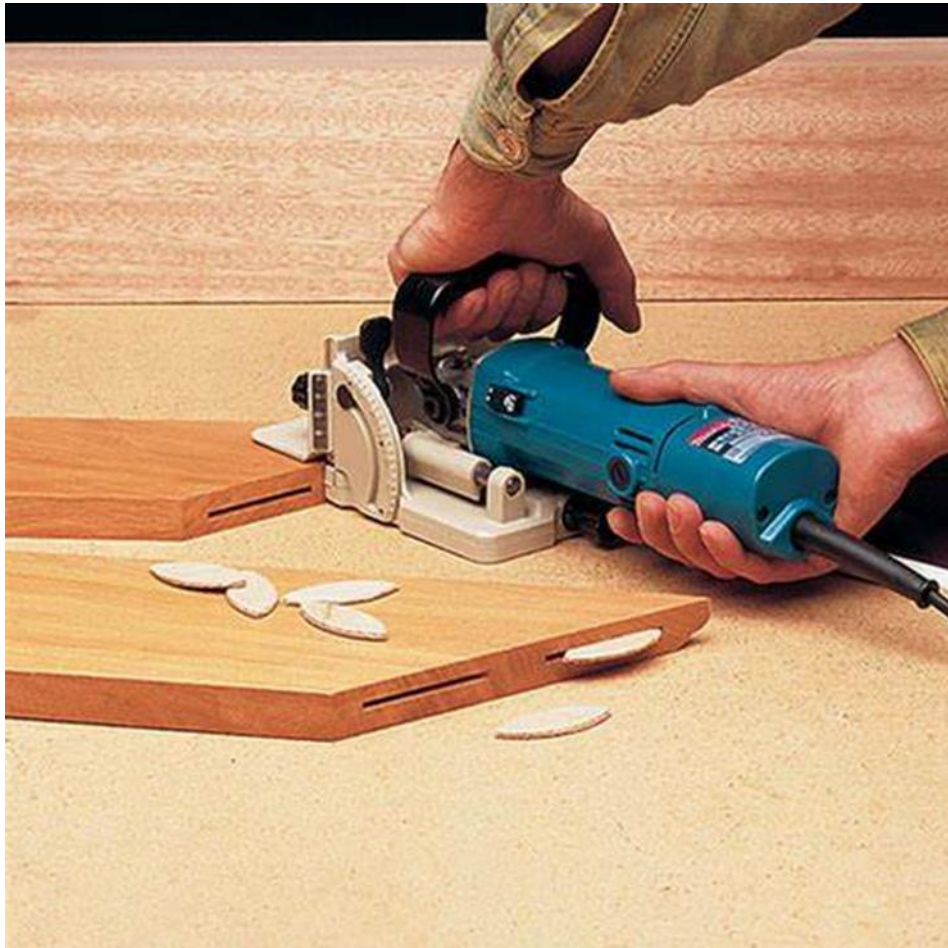


SLIP TONGUE & GROOVE JOINT
©FinePowerTools.com

DOWELLED JOINT



Biscuit joint



TERM 3

CUTTING LIST

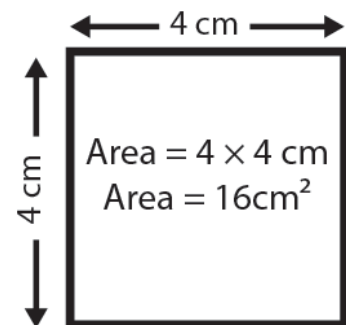
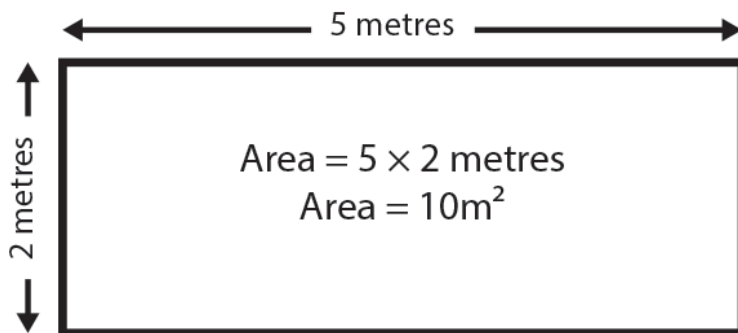
What is a cutting list?

A cutting list is a *list* of wood boards or panels *needed to build your product*.

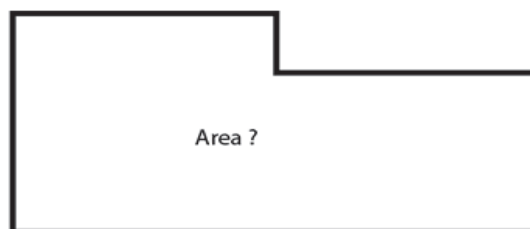
MEASURING

To measure Area of square and rectangle, you need to use the formulae for Area =

Length $\times$ Breathe = Area



Area of an odd shape



Solution 1



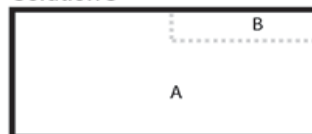
Total Area = Area A + Area B

Solution 2



Total Area = Area A + Area B

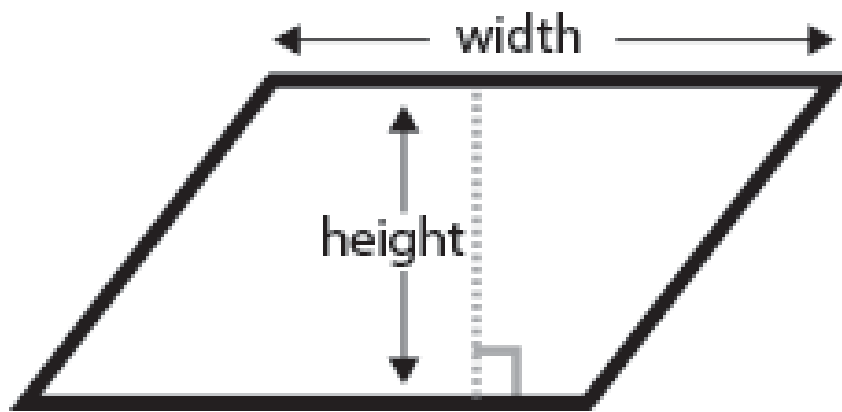
Solution 3



Total Area = Area A - Area B

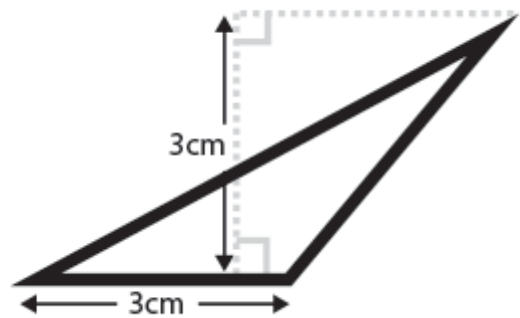
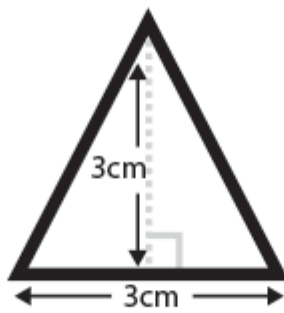
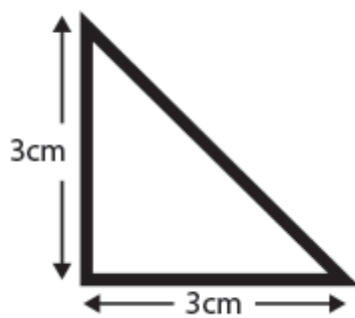
Area of Parallelogram

Width $\times$ Height = Area



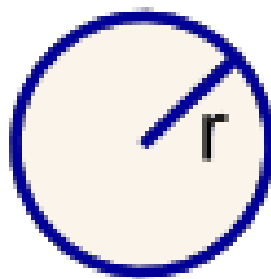
Area of a Triangle

$$\text{Height} \times \text{Base} = \text{Area}$$



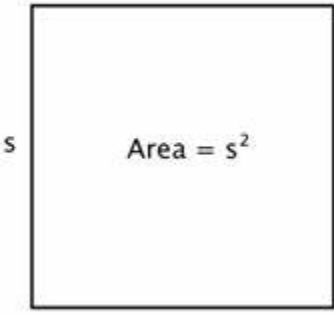
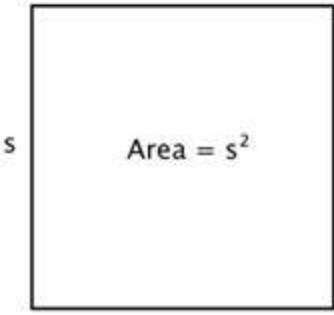
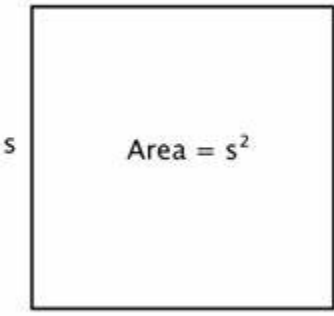
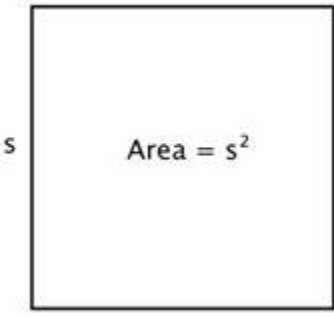
Area of a Circle

Area of a circle = πR^2 .



AREA CALCULATIONS

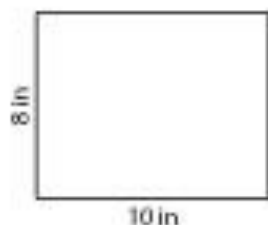
SQUARE

Shape		Calculate
		Side = 4cm
		Side = 5cm
		Side = 6cm
		Side = 70mm

Area of a Rectangle

Find the area of each rectangle.

1)



Area = _____

2)



Area = _____

3)



Area = _____

4)



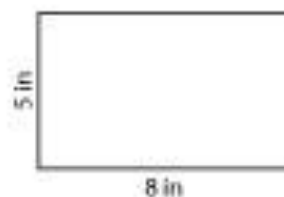
Area = _____

5)



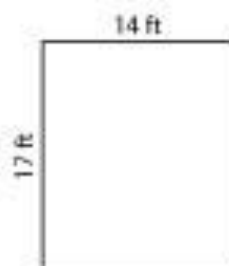
Area = _____

6)



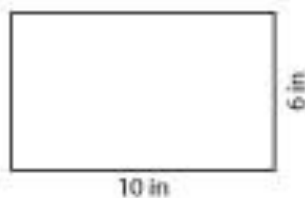
Area = _____

7)



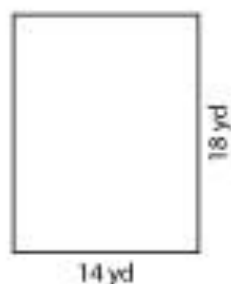
Area = _____

8)



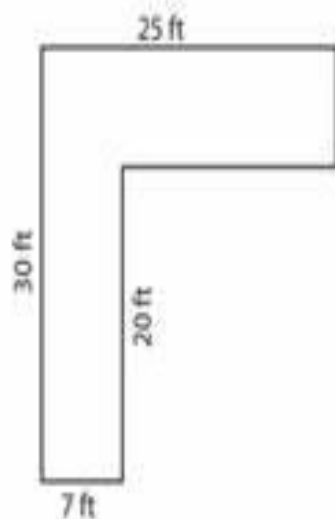
Area = _____

9)



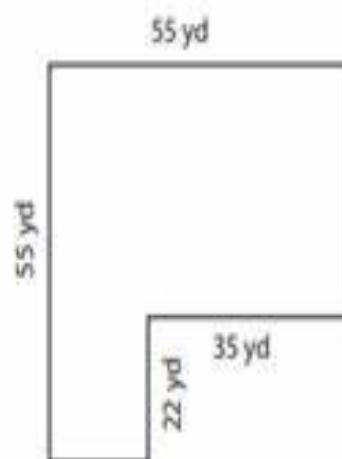
Area = _____

1)



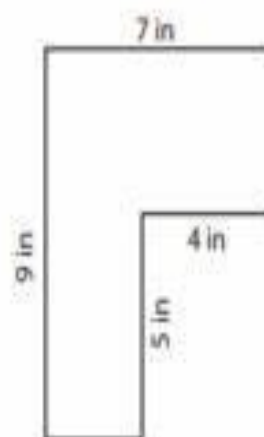
Area = _____

2)



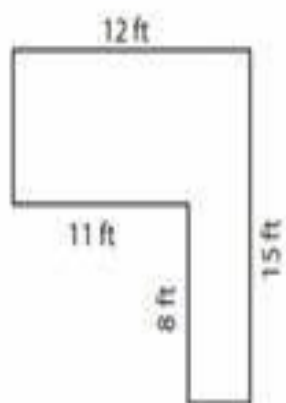
Area = _____

3)

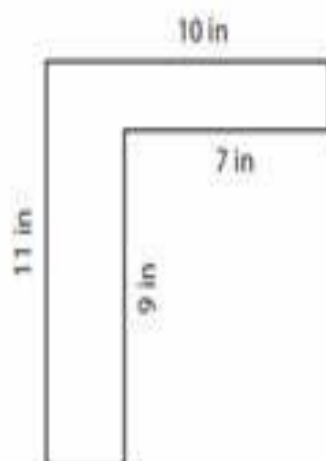


Area = _____

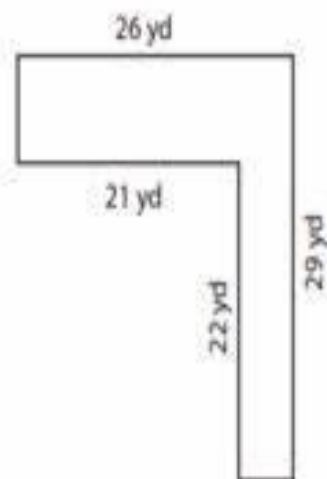
4)



5)



6)



Name _____

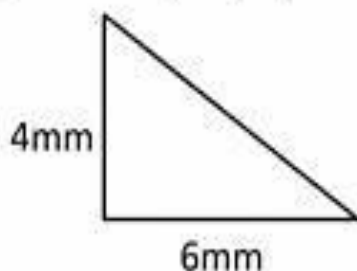
Date _____



TRIANGLE AREA SHEET 2

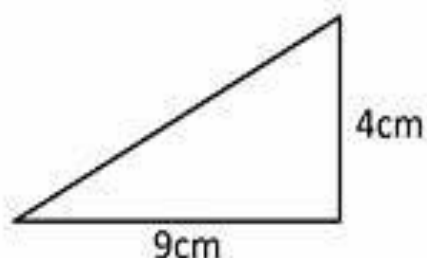
Work out the area of the following right angle triangles by halving the area of the rectangle formed by its perpendicular sides. They are not drawn to scale.

1)



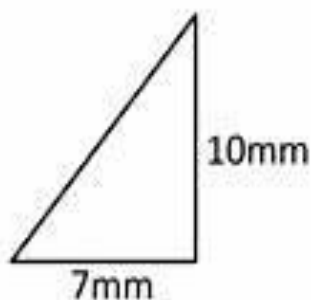
Area = _____ mm²

2)



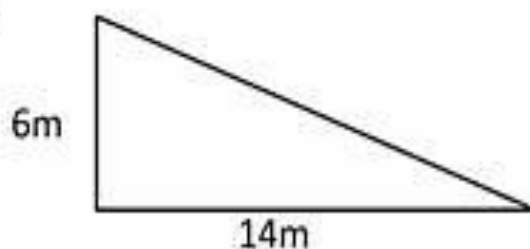
Area = _____ cm²

3)



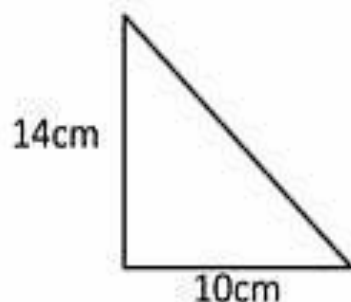
Area = _____ mm²

4)



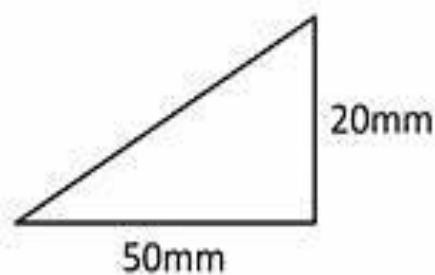
Area = _____ m²

5)



Area = _____ cm²

6)



Area = _____ mm²

Handy hint:

The formula for the area of a triangle is $\frac{1}{2} \times \text{base} \times (\text{perpendicular}) \text{ height}$



Free Math Sheets, Math Games and Math Help

MATH-SALAMANDERS.COM

Name : _____

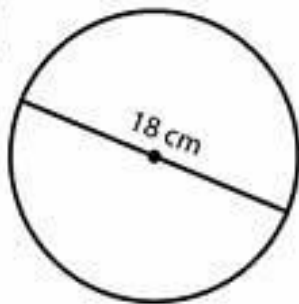
Score : _____ Date : _____



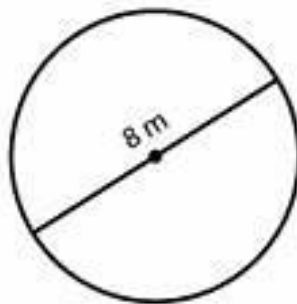
Area of a Circle

Find the area of each circle. Use $\pi = 3.14$

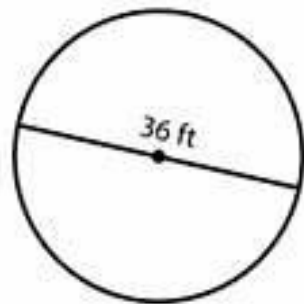
1



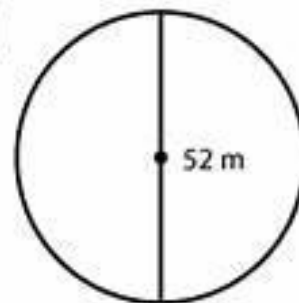
2



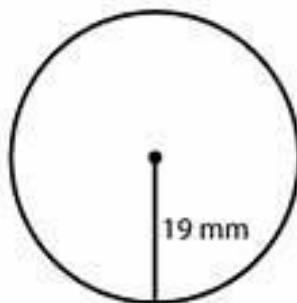
3



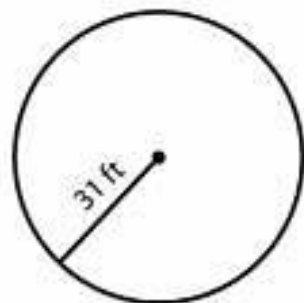
4



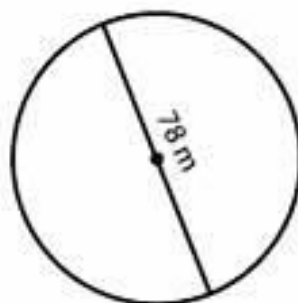
5



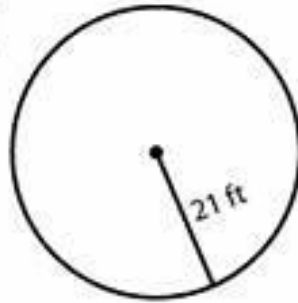
6



7



8



MATERIAL

SOFT WOOD

Types of Softwood

- Pine: easy to obtain and use for various applications, such as furniture, plywoods, fibre-boards, and construction framing.
- Cedar: a premium softwood that is denser, tougher, and more durable; mainly used for making outdoor furniture, decking, and other exterior projects.
- Spruce: used throughout construction, panelling and cladding.
- Larch: used for cladding and boats.
- Douglas-fir: used for joinery, doors and heavy construction.

Types of hardwood

1. **Alder:** Used for lightweight lumber with a wide-spaced wood grain pattern.
2. **Black walnut:** Produces expensive lumber with a rich, deep brown color.
3. **Brazilian cherry:** Used for various woodworking projects.
4. **Exotic expensive Furniture:** The most common use of hardwoods.
5. **Flooring and decking:** Highly preferred for aesthetics.

Softwood is light and hardwood is heavy

Softwood is easier to work with.

Softwood cuts easier than hardwood

Hardwood lasts longer than Softwood

Softwood bends easier than Hardwood

Softwood grows faster and easier than Hardwood

TICK THE CORRECT ANSWER

QUESTION	SOFTWOOD	HARDWOOD
Which wood is light		
Which wood is heavy		
Which wood is easy to work with		
Which wood cuts easy		
Which wood bends easily		
Which wood bends easy		
Which wood grows faster		

BOARDS

1. **Plywood:**

- **Composition:** Plywood consists of thin layers (plies) of wood veneer glued together with the grain direction alternating between adjacent layers. It typically has an odd number of layers.
- **Characteristics:**
 - Strong and durable.
 - Resistant to warping and splitting.
 - Available in various thicknesses.
- **Uses:**
 - Furniture construction.
 - Cabinetry.
 - Flooring.
 - Exterior sheathing.
 - Decorative panels.

2. **MDF (Medium-Density Fibreboard):**

- **Composition:** MDF is made from wood fibres and resin, compressed under high pressure.
- **Characteristics:**
 - Smooth surface, ideal for painting.
 - Uniform density.
 - Not as strong as plywood.
- **Uses:**

- Cabinet doors.
- Shelves.
- Decorative mouldings.
- Indoor furniture.
- Speaker enclosures.

3. **HDF (High-Density Fibreboard):**

- **Composition:** Similar to MDF but denser.
- **Characteristics:**
 - Even smoother surface.
 - Higher density.
 - Suitable for intricate designs.
- **Uses:**
 - High-quality furniture.
 - Decorative wall panels.
 - Crafts and art projects.

4. **Particleboard** (also known as chipboard):

- **Composition:** Made from wood particles bonded with adhesive.
- **Characteristics:**
 - Economical.
 - Less durable than plywood or MDF.
 - Commonly used in low-cost furniture.
- **Uses:**
 - Budget-friendly furniture.
 - Back panels of cabinets.
 - Underlayment.

5. **OSB (Oriented Strand Board):**

- **Composition:** Compressed wood strands arranged in layers.
- **Characteristics:**
 - Rough texture.
 - Strong and moisture-resistant.
 - Used primarily for structural purposes.

○

- **Uses:**

- Roof and wall sheathing.
- Subflooring.
- Outdoor applications.

6. **Veneer Boards:**

- **Composition:** Thin wood veneer glued onto a core material (usually MDF or particleboard).
- **Characteristics:**
 - Real wood surface.
 - Lightweight.
 - Affordable.
- **Uses:**
 - Cabinet doors.
 - Furniture surfaces.
 - Wall panels.

HINGES

Hinges

- Butt hinge
- Rising butt hinge
- Counter flap or butterfly hinge
- Parliament hinge
- Piano hinge
- Strap hinge
- T-hinge
- Band and hook hinge
- Sinkless hinge
- Concealed hinge
-

ACTIVITY

PICTURE	NAME	USE
	Butt hinge	Mostly used on house doors



**Butterfly
hinge**

**Mostly used
for cabinets**



**Rising butt
hinge**

**Used for
self-closing
doors**



**Parliament
hinge**

**Wide door
swing**



Piano hinge

**Used to
support lids
of boxes
and piano's**



**STRAP
HINGE**

**Used on
gates and
heavy doors**



T-hinge

**Used for
heavy doors
and gates
hung**



**Sinkless
hinge**

**To use for
cabinets un-
der a sink**



**Concealed
hinge**

**Used for
cupboard
doors**

PICTURE	NAME	USE
	Butt hinge	Mostly used on doors
	Butterfly hinge	Mostly used for cabinets
	Rising butt hinge	Used for doors



**Parliament
hinge**

Wide door

—



_ hinge

**Used to sup-
port lids of
boxes and pi-
ano's**



_ HINGE

**Used on gates
and heavy
doors**



_____-hinge

**Used for heavy
doors and
gates hung**



Sinkless hinge

To use for
cabinets
_____ a sink



**Concealed
hinge**

Used for

doors

LOCKS

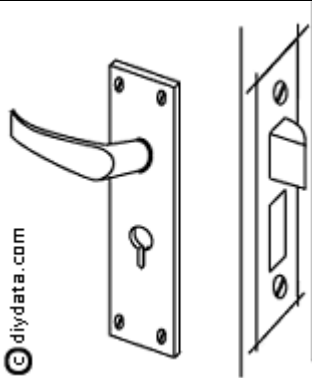
- Straight cupboard lock
- Box or kist lock
- Drawer lock
- Mortise lock
- Rim lock
- Night latch
- Sliding door lock

LOCK	NAME	USE
	Sliding door lock	Used on doors and slides into locks
	Straight cupboard locks	Light lock for cupboard doors
	Box locks	Used to lock gates and lockers



Drawer locks

Used for locking drawers



Mortise lock

Used inside the door



Rim lock

Used on front doors and it uses keys

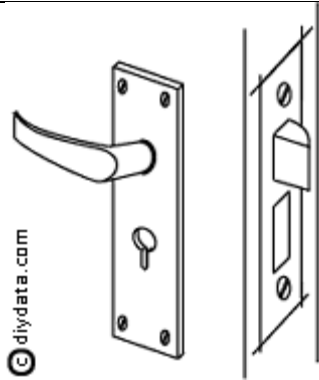


Night latch

Used on front doors without keys

ACTIVITY

LOCK	NAME	USE
	Sliding door lock	Used on doors and _____ into locks
	Straight cup-board locks	_____ lock for cup-board doors
	_____ locks	Used to lock gates and lockers
	Drawer locks	Used for locking _____



Mortise lock

Used

the door



Rim lock

**Used on front
doors and it
uses**



Night latch

**Used on front
doors**

keys